

Complete System Architecture Layers

Table 1. Repository-verified current-state layers of the ZENTRA / Zion Events Place system. Components are listed only where implementation or deployment configuration exists.

Architecture classification

Layered modular monolith with managed cloud services and event-driven integrations. The system contains **14 implemented architectural layers**. A future admin mobile application would reuse the same secured API and identity layers; it is not represented as an existing application.

NO.	SYSTEM LAYER	ACTUAL COMPONENTS	PRIMARY RESPONSIBILITY
1	Access Channel Layer	Guest/client browser; Admin/SuperAdmin browser; client verification session. The planned admin mobile application is not implemented.	Provides the authorized entry points used to access the system.
2	Presentation Layer	Next.js App Router pages, React components, responsive Tailwind layouts, public website, booking interface, and administrative dashboard.	Presents information and captures user interactions across supported web screen sizes.
3	Edge and Routing Layer	Next.js proxy, route matching, client package URL rewrite, and required-password-change redirect.	Processes selected routes before they reach the presentation or API layers. It is not the primary authorization boundary.
4	Identity and Access Layer	NextAuth credentials, JWT sessions, bcrypt password verification, database revalidation, requireAdmin, requireSuperAdmin, and client OTP grants.	Authenticates users, validates current account state, enforces roles, and protects private booking records.
5	API and Application Interface Layer	Route handlers, server actions, public APIs, protected admin APIs, and protected orchestration APIs.	Receives requests, validates interface-level input, applies access checks, and invokes application services.
6	Domain and Business Logic Layer	Booking, inquiry, calendar, payment, contract, services/packages, testimony, support, team, reports, task templates, and Command Center services.	Implements business rules, validation, state transitions, conflict handling, and transactional operations.
7	Workflow and Orchestration Layer	Booking orchestration services, n8n webhooks and callbacks, Command Center jobs, job leases, retries, idempotency controls, tasks, and notifications.	Coordinates asynchronous workflows while keeping authoritative business state inside the application database.
8	AI and Knowledge Layer	Smart Assistant, Gemini provider, permitted tool set, published knowledge index, pgvector search, full-text search, citations, and deterministic fallback.	Produces grounded responses from approved knowledge and permitted live records without granting the model unrestricted database access.
9	Data Access Layer	Prisma 7, PrismaPg adapter, server-side transactions, Supavisor transaction pooling, and direct migration configuration.	Provides controlled application access to persisted data and enforces transactional data operations.

NO.	SYSTEM LAYER	ACTUAL COMPONENTS	PRIMARY RESPONSIBILITY
10	Data Persistence Layer	Supabase PostgreSQL, pgvector, business tables, user records, jobs, settings, histories, timelines, and operational logs.	Acts as the single authoritative system of record for application and operational data.
11	Object Storage Layer	Supabase Storage REST adapter, private signed URLs, controlled public URLs, payment proofs, contracts, documents, testimony images, and media assets.	Stores binary files separately from relational records while preserving controlled server-side access.
12	Communication and External Integration Layer	Render-hosted n8n, Gmail API/OAuth, Gemini API, team/client access webhooks, booking webhooks, and contract delivery webhooks.	Connects the application to workflow automation, email delivery, one-time code delivery, and external AI processing.
13	Observability and Governance Layer	Audit logs, booking and contract timelines, payment histories, email logs, n8n workflow logs, assistant interaction logs, health checks, and request-error instrumentation.	Provides traceable evidence of user activity, business changes, automated effects, failures, and system health.
14	Infrastructure and Deployment Layer	Vercel Next.js runtime, Supabase PostgreSQL and Storage, Render n8n service, separate Render PostgreSQL for n8n state, Gmail, and Gemini.	Hosts and connects the application runtime, authoritative data services, storage, workflow engine, workflow-state database, email provider, and AI provider.

Strict layer boundary

Browsers and any future mobile client must use the Next.js interfaces and secured APIs. They must not connect directly to PostgreSQL, Supabase service-role storage, n8n shared secrets, Gmail OAuth credentials, or Gemini API keys. Supabase PostgreSQL remains the authoritative application system of record; Render PostgreSQL stores only n8n internal state.